Actor-Centric Turing Ordering Visualizer Implementation Note

April 28, 2026

Purpose

This note documents the *actor-centric* visualizer and ranking variant that was implemented in the `llm-eval` repository before the interface was switched back to the strict-ε representation. The point of the note is archival clarity: it states exactly which quantities were computed, how the graph was built, and how the ranking was derived from the same pairwise score.

1. Primitive empirical quantities

For each ordered pair (A, B),

- A is the actor,
- B is the distinguisher / target.

The implementation used two empirical frequencies:

$$s_B := \Pr(B \text{ answers “same”} \mid B \text{ vs } B),$$

$$r_{A\to B} := \Pr(B \text{ answers “same”} \mid A \text{ imitates } B).$$

Operationally,

- s_B is the self-recognition rate of the judge B;
- $r_{A\to B}$ is the fooling rate of the actor A against B.

The original strict visualizer also used

$$P_{\text{correct}}(A, B) = \frac{1}{2}\Pr(y = 1 \mid B, B) \; + \; \frac{1}{2}\Pr(y = 0 \mid A, B).$$

Since $\Pr(y = 1 \mid B, B) = s_B$ and $\Pr(y = 0 \mid A, B) = 1 - r_{A\to B}$, this becomes

$$P_{\text{correct}}(A, B) = \frac{1}{2}s_B + \frac{1}{2}(1 - r_{A\to B}).$$

2. Actor-centric pairwise margin

The actor-centric implementation was based on the margin

$$m(A, B) := \frac{1}{2}(r_{A \to B} - s_B).$$

Equivalently,

$$m(A, B) = \frac{1}{2} - P_{\text{correct}}(A, B).$$

Hence:

- $m(A, B) > 0$: actor A fools B better than B recognizes a genuine B;
- $m(A, B) = 0$: actor performance matches the judge's self-recognition baseline;
- $m(A, B) < 0$: the judge still distinguishes better than this actor fools.

This was called *actor-centric*, not because it removed the judge from the formula, but because the sign convention made larger values correspond to stronger actor-side fooling.

3. Graph rule

The actor-centric graph replaced the strict-ε rule with a direct threshold on m. For a user-chosen threshold $\tau \geq 0$, the graph displayed the directed edge

$$A \to_\tau B \quad \iff \quad m(A, B) \geq \tau.$$

Interpretation:

- larger τ makes the graph stricter;
- an edge survives only when the actor-side fooling surplus is large enough;
- the edge label shown in the UI was $m(A, B)$.

This produced a graph whose semantics were pairwise and actor-focused:

"A fools B by at least τ".

4. Ranking

The ranking was intentionally derived from the same pairwise margin, so that the graph and the ranking used one coherent notion of actor-side performance.

4.1 Main score

The primary score was

$$A^+(A) := \operatorname*{mean}_{B \neq A} \max\{0, m(A, B)\}.$$

This is the positive-part average of the actor-centric margin across all targets. It can be read as:

- how often the actor is above the baseline at all;
- how strongly it exceeds that baseline when it does;
- an area-under-threshold summary of the same edge rule used by the graph.

4.2 Auxiliary summaries

Two additional summaries were displayed:

$$\bar{m}(A) := \operatorname*{mean}_{B \neq A} m(A, B),$$

$$\mathrm{out}_\tau(A) := \#\{B \neq A : m(A, B) \geq \tau\}.$$

So the ranking table exposed:

- $A^+(A)$: positive-part actor score;
- $\bar{m}(A)$: signed mean actor margin;
- $\mathrm{out}_\tau(A)$: current graph out-degree.

5. Circular ordering of nodes

The graph layout put all models on a circle. The nodes were ordered clockwise by the primary ranking score A^+, from strongest fooler to weakest. Therefore:

- the *graph* showed who fools whom at threshold τ;
- the *circular order* showed the global actor ranking;
- the *edge labels* showed the exact pairwise margin $m(A, B)$.

6. Data pipeline

The actor-centric implementation consumed the same pairwise evidence objects as the strict visualizer. For each pair row:

1. recover s_B from the self branch;
2. recover $r_{A \to B}$ from the imitation branch;
3. compute $m(A, B) = \frac{1}{2}(r_{A \to B} - s_B)$;
4. draw the edge if $m(A, B) \geq \tau$;
5. aggregate $m(A, B)$ over targets to produce $A^+(A)$ and $\bar{m}(A)$.

7. Difference from the strict-ε visualizer

The strict visualizer is judge-centric:

$$d(A, B) = P_{\mathrm{correct}}(A, B) - \frac{1}{2}, \qquad A \to B \iff d(A, B) \leq \varepsilon.$$

The actor-centric visualizer in this note instead used

$$m(A, B) = \frac{1}{2} - P_{\mathrm{correct}}(A, B), \qquad A \to B \iff m(A, B) \geq \tau.$$

The two are equivalent under the identity

$$m(A, B) = -d(A, B),$$

but they emphasize opposite sides of the same empirical relation:

- d measures judge advantage above chance;
- m measures actor surplus above the judge's self-recognition baseline.

8. Status

This actor-centric formulation is preserved here as an implementation record. The currently restored visualizer in the repository returns to the strict-ε graph and the earlier $S/A/J$ ranking family, but this note remains useful because it explains the alternate actor-centric branch precisely and unambiguously.